

DIPOLE DETECTOR — BUILD SHEET

An instrument, 3 variants: DD-433 (Exp. B, 433.59 MHz M&W) · DD-1296 (Exp. A, 1296 MHz) · DD-2450 (2.45 GHz oven test)

1. CUT LENGTHS (1/8 in = 3.175 mm brass rod, 1.5 mm feed gap)

	DD-433 433.59 MHz	DD-1296 1296 MHz	DD-2450 2450 MHz
Wavelength	69.1 cm	23.1 cm	12.2 cm
CUT at (tip to tip, long)	345.7 mm (13.61 in)	115.7 mm (4.55 in)	61.2 mm (2.41 in)
each half at cut	172.1 mm (6.78 in)	57.1 mm (2.25 in)	29.8 mm (1.17 in)
EXPECTED final (tip to tip)	332.2 mm (13.08 in)	109.6 mm (4.32 in)	57.2 mm (2.25 in)
each half, final	165.4 mm (6.51 in)	54.1 mm (2.13 in)	27.9 mm (1.10 in)
trim per tip (about)	6.7 mm	3.0 mm	2.0 mm
Balun sleeve, start (VF 1.00)	172.9 mm (6.81 in)	57.8 mm (2.28 in)	30.6 mm (1.20 in)
Balun sleeve if taped (VF 0.95)	164.2 mm (6.47 in)	54.9 mm (2.16 in)	29.1 mm (1.14 in)

Half = (tip-to-tip – 1.5 mm gap) ÷ 2, measured from the gap end of the rod to its tip. CUT LONG, then trim both tips equally in small steps (0.5 mm), checking resonance with a VNA, or TinySA + generator + return-loss bridge. Aim for the lowest SWR at the band. Expected final = free-space half-wave × thickness factor K (DD-433 0.961, DD-1296 0.948, DD-2450 0.935), read from the classical chart RSGB Handbook 2011 Fig 16.1 (G8EZE). Published charts disagree slightly, so TRIM, don't trust. A computer check (openEMS) agreed the thick rod is ~3–4 % shorter than thin wire at 2450 MHz. Inside the cage the resonance moves: do the final trim IN THE CAGE.

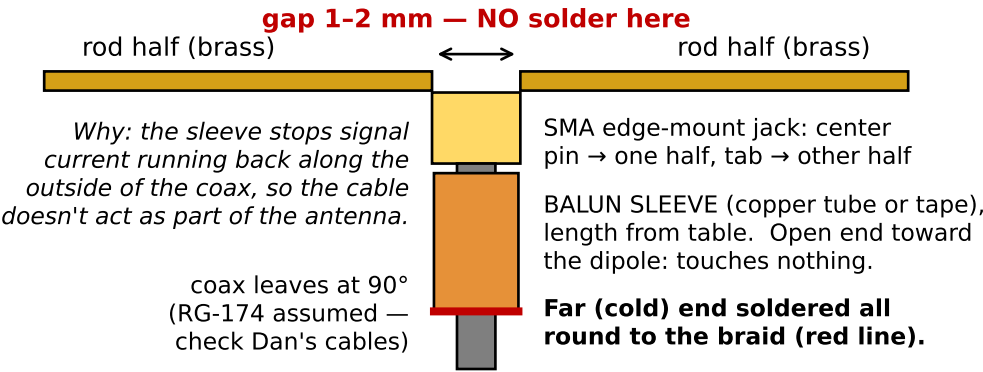
WHY 1/8 in BRASS IS FINE FOR ALL THREE

- Stiff: holds its shape without a support.
- Solders easily to the SMA jack.
- Thickness only makes the cut length a few % shorter (already in the table above).
- A thicker rod gives a broader, easier-to-hit tuning range than thin wire.
- Brass is chosen for its electrical conductivity (about 3× tungsten W1) — nothing else.

PARTS (all 3 variants; see PARTS_LIST.md)

- Brass rod 1/8 in: 523 mm total cut → buy 2 × 36 in or 5 × 12 in (generic)
- SMA edge-mount jack Amphenol 132322 (Digi-Key ACX1652-ND): 3 + 2 spares
- 3 SMA coax pigtails ~30 cm (RG-174 assumed, or Dan's existing assemblies)
- Thin-wall copper/brass tube or copper tape, heat shrink, nylon clamps, HDPE plate

2. FEED + BALUN (top view, not to scale)

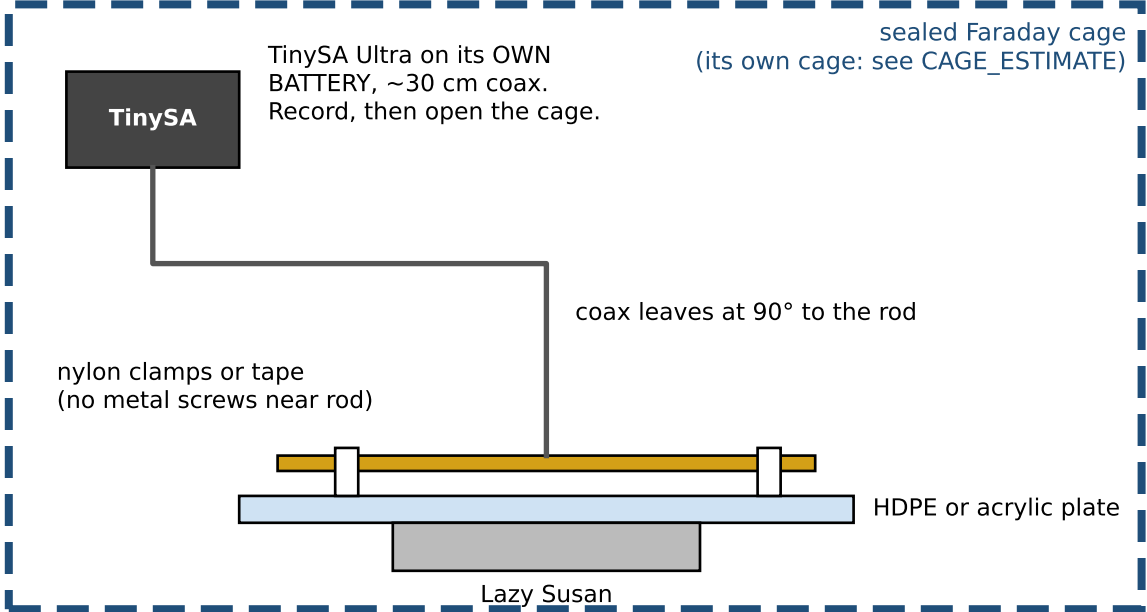


3. HOW TO SOLDER

1. Cut halves LONG (table). File ends square.
2. Tin the gap end of each rod and the SMA pin + tab.
3. Clamp both halves on a straight edge: in line, gap 1–2 mm.
4. Solder one half to the center pin, the other to the ground tab. Small, quick joints. NO solder bridging the gap.
5. Check tab geometry on the Amphenol 132322 datasheet first.
6. Balun: 2 layers tape over jacket → slide sleeve on → open end at the jack → solder FAR end to braid → heat-shrink.

THIS IS A RECEIVING ANTENNA,
NOT a Monstein & Wesley passive polarizer rod.
It replaces one rod in the array for detection only.

4. MOUNTING + CAGE (side view)



NO CABLE CROSSES THE CAGE WALL. No feed-through, no bulkhead.

WHAT THIS MEASURES

HYPOTHESIS (SLW): the dipole couples to the longitudinal field through charge build-up (Hively, $J = \sigma E$; US 9,306,527), and the TinySA reads the signal induced at the feed gap.

ORDINARY RADIO (classical fact): any dipole also picks up normal radio waves. The closed cage is there to block those. Whether the SLW gets through the cage wall is a HYPOTHESIS being tested.

Brass is specified for its conductivity (σ) only.

SAFETY: detectors, rods and TinySA stay OUTSIDE the oven cavity.
Never put anything inside a running microwave oven.

Dipole Detector DD-433 / DD-1296 / DD-2450

Grok Bot initial design — VERIFY BEFORE CUTTING

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